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A GCN-based adaptive generative adversarial network model for short-term wind speed scenario prediction

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ABSTRACT

Wind prediction is of great significance for wind energy utilization due to the stochastic nature of wind. To effectively facilitate various downstream decision-making tasks such as wind turbine control, predictive wind Scenario Generation (SG), which is capable of providing a set of deterministic instantiated wind prediction results, plays a critical role. In this paper, a novel Graph neural networks-based Adaptive Predictive Generative Adversarial Network (GAPGAN) model is proposed for accurate prediction of short-term future scenarios of a wind field. In GAPGAN, the original multivariate time series data are first reconstructed into the form of a graph, and spatiotemporal features are then extracted using Graph Convolutional Networks (GCNs). Next, a predictive generative adversarial network (PGAN) framework is proposed, which could generate different outputs corresponding to given historical observations as conditions. Finally, an adaptive PGAN training mechanism is introduced to stabilize the training process, and the best SG model is selected based on the proposed comprehensive evaluation system. Based on wind speed data collected from 11 wind turbines, computational experiments validate that the GAPGAN outperforms five benchmarking models in terms of point prediction accuracy, shape similarity, uncertainty prediction quality, and prediction scenario diversity.

1. Introduction

In light of the severe situation of climate and environment change, wind energy has been increasingly exploited and developed for its clean and sustainable advantages. However, wind conditions exhibit strong randomness and intermittency, causing significant interference to wind power generations and affecting the reliability of the power grid [1,2]. Thus, accurate wind predictions for wind farm operators is crucial for decision-making and risk management [3]. Existing wind speed prediction techniques can be categorized into three classes based on the form of the prediction results: point prediction, interval prediction, and scenario generation (SG), which are visualized in Fig. 1 to demonstrate their differences.

Point predictions are the most straightforward and simplest approach to provide wind farm operators with single-value wind speed predictions, and as such have been most widely studied in the previous literature. However, despite the simplicity and clarity of this approach, since there always exist prediction errors, single-value predictions fail to reflect the prediction uncertainty and limit the effectiveness of risk management [4–6].

Compared to point predictions, interval predictions are more effective in assessing the uncertainty of wind prediction results by generating lower-upper bounds. Despite the ability to depict the prediction uncertainty, the upper and lower bound prediction results are difficult to apply to decision-making models, which are often represented as stochastic optimization models [7,8]. As a result, the practicality of the interval prediction results is still limited.

SG is the most up-to-date wind prediction approach since recent years due to the rapid rise and development of generative models in the artificial intelligence domain. In the considered wind prediction problem, each *scenario* can be regarded as a single possible outcome, i.e. an instantiated future wind condition. Thus, SGs for wind prediction aim to create multiple plausible future wind conditions based on historical wind data. Based on a set of deterministic instantiated prediction results, SG can facilitate to predict the uncertainty and main trend of the future wind field simultaneously, and the generated results can be easily used in almost any form of decision-making models compared to interval prediction results [9]. In general, point predictions have inherent

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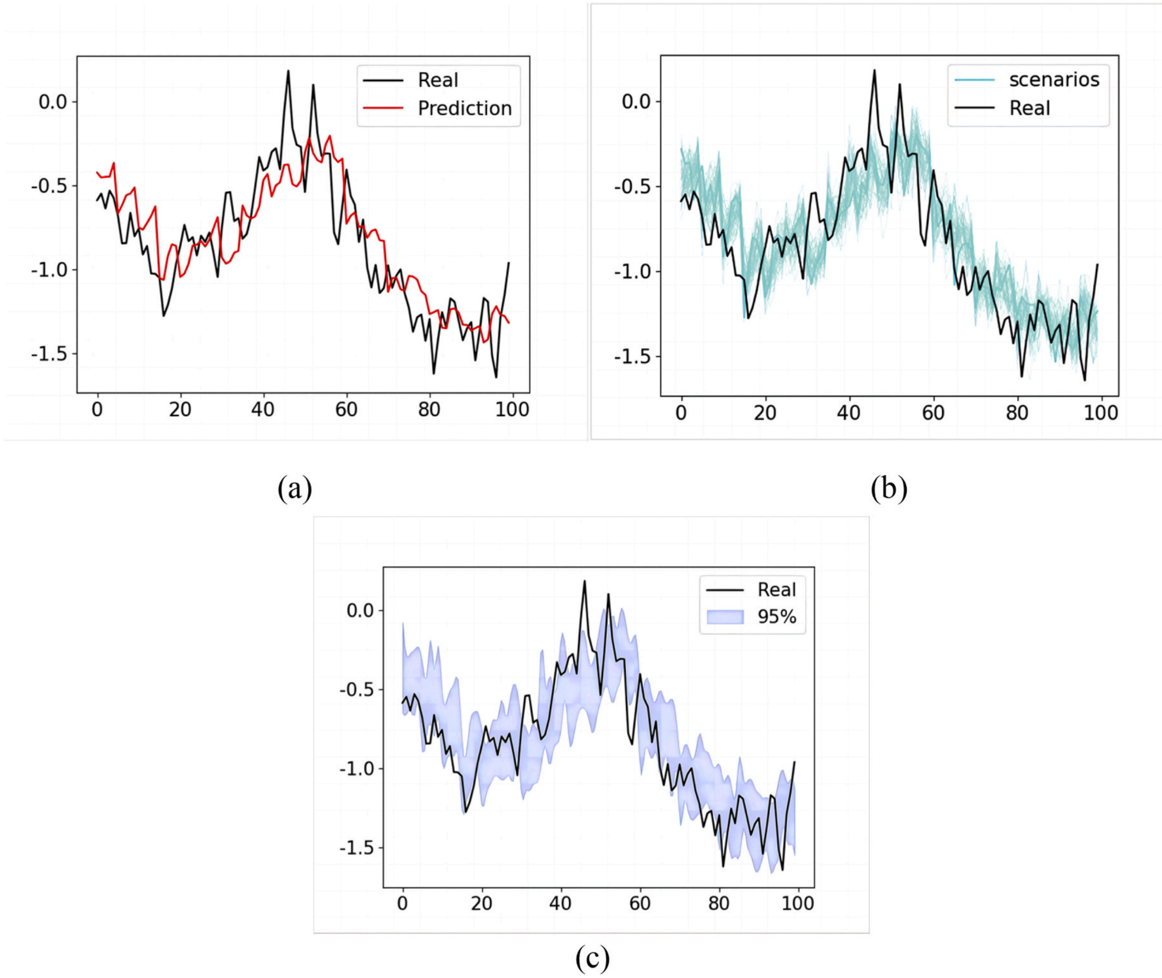


Fig. 1. Visualization of three different wind power prediction schemes: (a) point prediction, (b) scenario generation, and (c) interval prediction.

absolute errors, while interval predictions cannot provide sufficient information for subsequent decision-making. On the other hand, scenario generation enables uncertainty prediction for highly stochastic wind speeds by using instantiated predictive scenarios and the result can be used in many decision-making models such as stochastic optimization. Considering the advantages, we focus on SG-based wind speed prediction in this paper.

Recently, the advancement of deep learning has led to a variety of advanced models for SG. Among deep learning models, generative adversarial networks (GAN) have shown its powerful sample generation capabilities [10,11] and provide a meaningful framework for the wind SG problem. However, existing GAN-based SG models still present several limitations. First, wind field data exhibit strong spatiotemporal patterns and should be reasonably organized to extract features effectively. Traditionally, the wind field data can only be converted into the form of vectors or matrices to be processed by deep learning models, which restricts the representation flexibility of wind field data as well as the feature extraction effect. To address this issue, in Ref. [12], graph convolutional models (GCN) were employed for wind speed SG. However, in Ref. [13], the limitations of GCN models were pointed out, that is, the structure of GCN is more suitable for identifying useful structures in downstream tasks such as graph matching, graph classification, and graph clustering. Consequently, a standalone GCN model faces challenges in effectively extracting temporal features from multidimensional wind field data. In addition, GAN-based SG approaches rely on categorical labels for generating scenarios, which cannot provide better performance results in terms of prediction accuracy. Besides, GAN models are notoriously difficult to train [14], as the generator and discriminator are hard to converge in an adversarial training scheme. Also, since SG results can be used for multiple purposes and there exist many evaluation indicators accordingly, it is difficult to set a unified but versatile principle to select the optimal SG model. To address the issues above, the main contributions of this paper are as follows:

1. This paper proposes a novel GAN-based wind SG model, graph neural network (GNN)-based adaptive predictive GAN (GAPGAN) model, which is capable of mining spatiotemporal features of the wind effectively. To extract spatial features of a wind field, raw data are organized into the form of a graph, a more flexible representation than vectors or matrices, and spatial features are extracted via graph convolutions (GCs). Simultaneously, temporal convolutions complement this process by bridging the gap in feature extraction along the temporal dimension, a capability absent in GCs. The model intricately intertwines spatial and temporal feature processing modules, fostering a comprehensive understanding of spatiotemporal features in the wind data.
2. A novel GAN model, PGAN, for the predictive use of wind speed time series is developed, and a GAN training strategy is accordingly proposed to address the convergence imbalance problem between the generator and discriminator.
3. Considering the characteristic that SG results can be used for multiple purposes, including point prediction and interval prediction while ensuring statistical fidelity of wind speed patterns, a comprehensive SG evaluation system is designed by combining the evaluation metrics of point prediction, shape accuracy and uncertainty prediction, which can be used to select an optimal SG model.

This paper is structure as follows. In Section 2, related works are introduced. In Section 3, preliminary theories and models involved in this paper are introduced. In Section 4, the proposed framework for short-term wind predictive SG model, GAPGAN, is presented and explained. In Section 5, experimental studies are conducted based on real data to validate the proposed model from various aspects. In Section 6, conclusions are summarized and future work is introduced.

2. Related works

2.1. Review of existing wind speed prediction methods

Previous research has generally employed three types of methods to provide wind speed prediction: point prediction, interval prediction, and scenario generation, which are visualized in Fig. 1 to demonstrate their differences. First, point prediction aims to offer estimations of wind speed trends based on the analysis of historical wind speed. In Ref. [15], a novel information screening module, RTRD, was combined with a bidirectional long-short term memory (BiLSTM)-Attention module to predict wind speed. In Ref. [16], ensemble empirical mode decomposition (EEMD) was used to process wind speed data, and the multi-objective particle swarm optimization algorithm (MOPSO) was next applied to integrate multiple models to improve prediction accuracy. In Ref. [17], A novel ensemble model based on graph attention network (GAT) and GraphSAGE model was developed to predict wind speed using a bi-dimensional approach. Since point predictions cannot be perfectly accurate, their inherent errors should be accurately estimated.

Compared to point predictions, interval predictions provide a range of values instead of a single predictive outcome. In Ref. [18], the wavelet transform was first used to decompose the wind speed signal, and the sub-series obtained was next fed into a convolutional neural network (CNN) for point prediction. Finally, the interval prediction result was derived from the prediction errors assumed to follow a Gaussian distribution. In Ref. [19], the authors applied the kernel density estimation (KDE) method to predict and express the probability distribution of wind power. In Ref. [20], combining numerical weather prediction (NWP) data and historical wind data, an improved lower–upper bound approximation was applied to build reliable prediction intervals. Although interval predictions have advantages in uncertain prediction, their form poses challenges in providing support for subsequent wind turbine decision-making models.

SG can provide multiple instantiated possible outcomes in future for the decision-making models. Existing SG methods can be further categorized into two types: statistical method-based and machine learning-based SGs. The former requires to explicitly define the probability distribution of wind conditions and then do sampling from the distribution to generate a set of instances [21,22]. In Ref. [23], radial basis function neural networks (RBFNN) were combined with particle swarm optimization (PSO) algorithm to yield a SG model and generate accurate scenarios. Although statistical methods are more interpretable than machine learning models, this approach requires certain assumptions on data distributions, which are usually unknown, limiting the accuracy of SG results. More importantly, the statistical SG approach could only generate wind field scenarios for the general descriptive case and is difficult to use for predictive modeling.

In contrast, machine learning-based SG methods do not require assumptions about data distributions and can better characterize the stochastic nonlinear nature of wind patterns and facilitate predictive modeling. Particularly, deep learning, as the most popular and promising research area in machine learning, can effectively capture highly complex time series data patterns and demonstrates strong potential for the SG problem. In Ref. [24], a long-short term memory (LSTM)-based SG method was proposed to characterize the randomness of the electricity market. In Ref. [25], based on the Gaussian mixture model (GMM), an adaptive method for clustering wind field data was proposed, and an adaptive SG model was introduced based on LSTM network. In Ref. [26], an LSTM model was similarly implemented to predict the average wind direction and to form a wind direction predictor. Existing wind speed prediction studies are summarized and presented in Table 1.

Table 1
Review of wind speed prediction methods.

Author	Category	Models
[15] Ke L et al., 2022	Point prediction	RTRD + BiLSTM
[16] He X et al., 2020	Point prediction	EEMD + MSPSO
[17] Santos VO et al., 2023	Point prediction	GAT + GraphSAGE + LSTM
[18] Wang H et al., 2017	interval prediction	Gaussian + CNN
[19] Niu D et al., 2022	interval prediction	KDE + BiLSTM
[20] Wen P et al., 2020	interval prediction	KDE + CSSA
[21] Krishna A et al.,	Scenario generation	Vine Copula
[11] Zhang Y et al., 2020	Scenario generation	SVC + GAN
[12] Cho Y et al., 2023	Scenario generation	GCN + GAN

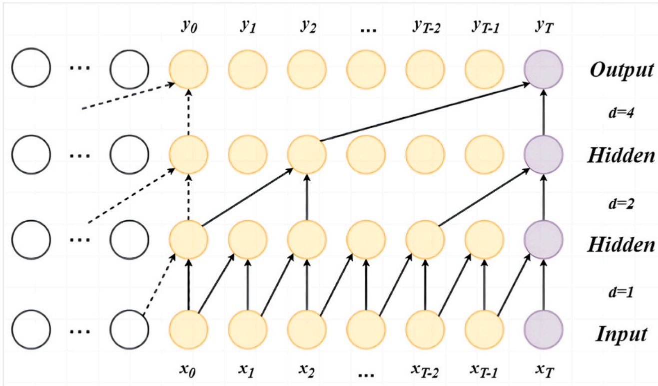
However, existing SG methods have limitations on prediction accuracy. First, these methods often focus on generating generic wind scenarios, which are not for the use of short-term predictions. It is necessary to associate generated scenarios with the latest observed wind speed data for higher short-term wind predictions. Besides, existing spatiotemporal feature extraction methods are not sufficiently accurate. A more advanced feature extraction method that deeply couples spatial and temporal features remains to be explored.

To enhance the SG accuracy, this paper integrates the principles of point prediction into SG and enables the generation of wind speed scenarios with short-term predictive features. Additionally, a novel spatiotemporal extraction method is proposed and applied in deep learning models.

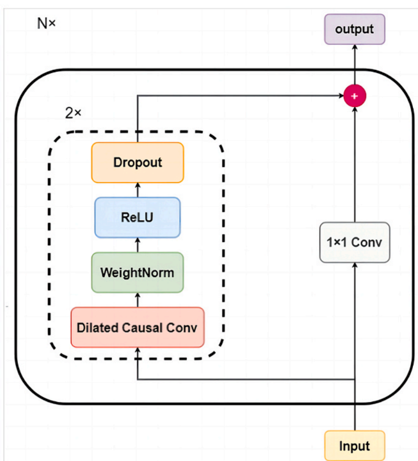
2.2. Potential of GAN models in SG

Recently, the booming development of computer science domain has driven the rapid growth of GAN modeling research and led to many advanced GAN variants, such as conditional generative adversarial networks (CGAN) and Wasserstein generative adversarial network (WGAN). CGAN, a type of conditional generative model [27], improves the original GAN by incorporating conditional information into both the generator and the discriminator to guide the generation process, ensuring that the generated output corresponds to the given condition. Besides, to eliminate the mode collapse problem of GAN models, WGAN as a type of generative adversarial network was proposed. Compared to the original GAN using Jensen-Shannon divergence or Kullback-Leibler divergence as the loss function, WGAN applying the Wasserstein distance as the loss function can ensure a stable and trainable gradient signal [28]. To apply GAN model to the wind energy industry, in Ref. [29], enforced Lipschitz constraint was used to alleviate overfitting in GAN, improving the quality of generated scenarios. In Ref. [11], a wind power SG framework based on the conditional improved WGAN was introduced, including cluster analysis, support vector classifier (SVC) for labeling, and conditional SG for modeling the uncertainty and variations in wind power.

So far, many researchers have attempted to utilize GAN models for wind speed SG. However, only a few researchers have addressed the issue of the challenging training process of GAN models in wind speed SG, and these GAN-based methods for wind speed SG have shown deficiencies in terms of prediction accuracy. To enhance the prediction accuracy and training efficiency of wind speed SG, this paper proposes a short-term Predictive Generative Adversarial Network (PGAN) specifically designed for short-term wind speed predictions. Additionally, an adaptive training approach is introduced to tackle the training difficulty of PGAN in wind speed SG.



(a)



(b)

Fig. 2. TCN model structure: (a) dilated causal convolutions with filter size $k = 2$ and dilation factors $d = 1, 2, 4$; (b) a typical network block in a TCN network.

3. Preliminaries

In this section, models involved in the proposed GAPGAN model, including temporal convolutional networks (TCN), GNN, and GAN, are introduced and explained to provide preliminary knowledge.

3.1. Temporal convolutional network

TCN, a popular family of deep learning architectures [30], has shown satisfactory results in processing time series data and performing predictions in recent years. Advantages of TCN include low data volume requirements, robust prediction results, and high prediction accuracy [31].

TCN models follow two principles: mining features without revealing future data, and generating corresponding outputs for each timestamped input data. To achieve these two principles, 1-D fully-convolutional network (FCN) [32] and dilated causal convolution are incorporated and implemented in TCN. As shown in Fig. 2(a), compared to traditional convolutional layers, dilated convolutional layers introduce fixed spacings between two neighboring elements to capture temporal relationships at various scales. Given an input sequence $x \in \mathbb{R}^n$ and a convolutional filter f_d with a dilation factor denoted by d, the dilated causal convolution operation F on an element of x indexed by s is expressed in (1)

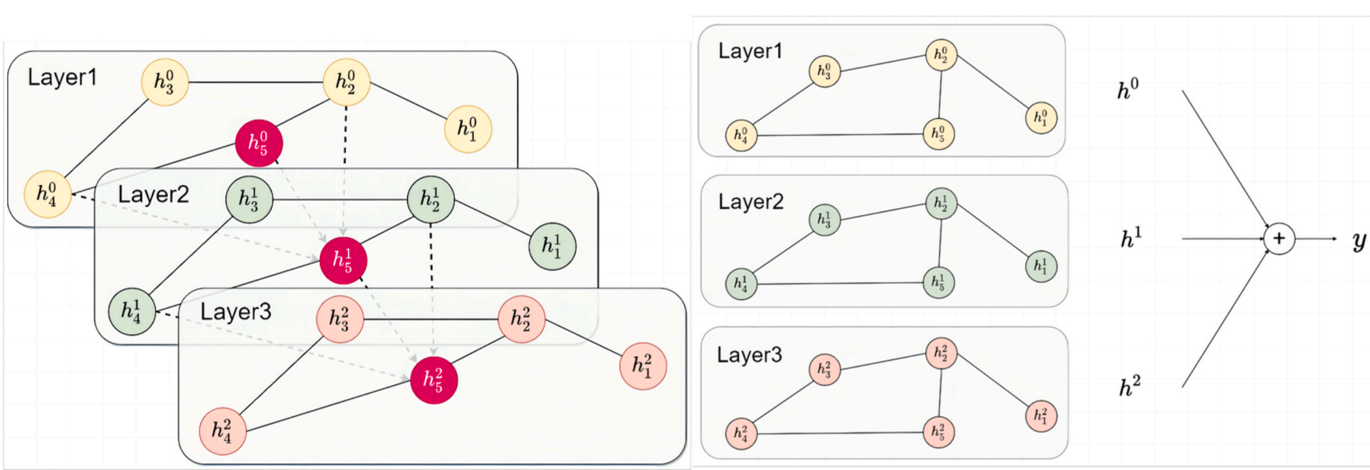
$$F(s) = (x * f_d)(s) = \sum_{i=0}^{k-1} f(i) \bullet x_{s-d\bullet i} \tag{1}$$

where k is the filter size, and $x_{s-d\bullet i}$ accounts for the observations before x_s. Next, as shown in Fig. 2(b), the outputs of the dilated convolutional layer and a convolutional layer are aggregated to form a block, which is further stacked one by one to construct the TCN model.

3.2. Graph neural networks

GNN, as a group of machine learning models for processing data of a graph structure, can effectively extract dependencies between nodes in a graph and update the states of nodes based on the information transmitted from their neighboring nodes at any depth. Generally, GNNs can be categorized into four classes: recurrent GNN, convolutional GNN, graph autoencoders, and spatiotemporal GNN [33]. In this paper, convolutional GNNs are used because of their advantages in extracting spatial relationships.

As shown in Fig. 3, in a convolutional GNN, the information of a node in the deep layer is first extracted based on the states of its adjacent nodes in shallow layers, and the final output is ultimately obtained by extracting the information contained in all layers. Operations of a GNN model are expressed in (2) and (3):



(a) (b)

Fig. 3. The computational process of a convolutional GNN: (a) the state updating process of a node, (b) the state prediction in convolutional GNNs.

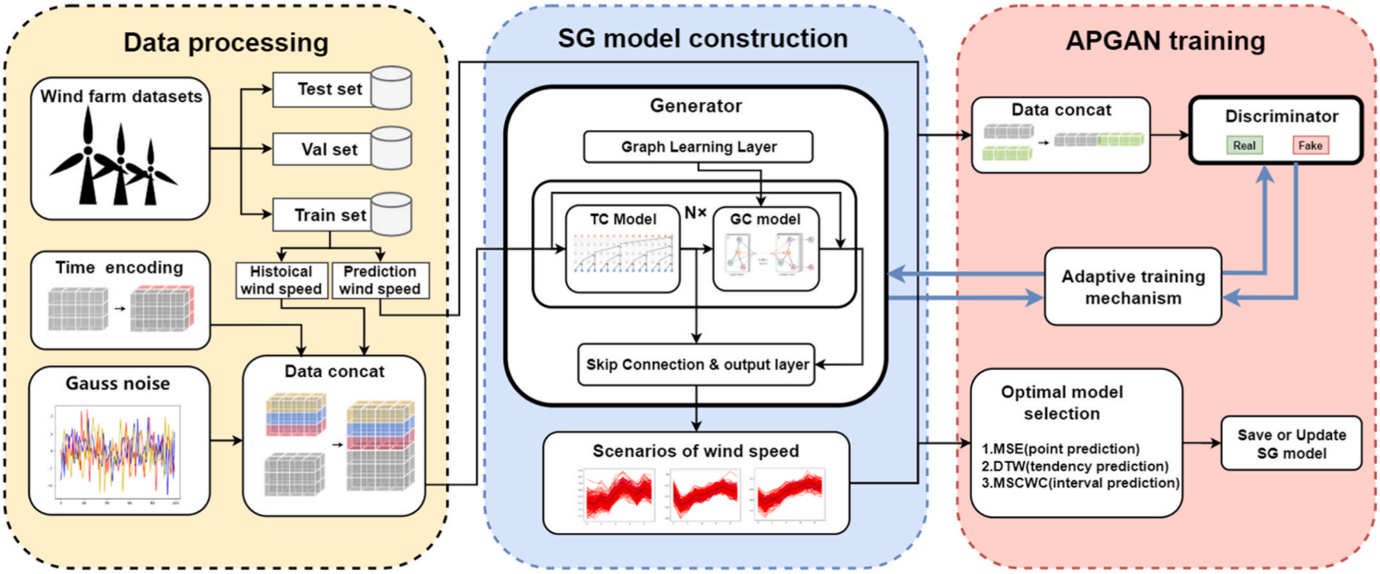


Fig. 4. The framework of the proposed GAPGAN model.

$$h_v^{t+1} = \sigma\left(\left(h_v^t + \sum_{u \in \mathscr{N}_v} h_u^t\right) \bullet W_{\mathscr{N}_v}^{t+1}\right) \tag{2}$$

$$y = \sum_{t=0}^{m} \omega_t \cdot h^t \tag{3}$$

where h_v^t denotes the state of node v in the layer t, h^t represents the state of all nodes in the layer t, $\mathscr{N}_v$ is the collection of nodes excluding the node v, σ is an activation function, m represents the maximum depth of the GCN, $W_{\mathscr{N}_v}^{t+1}$ and ω_t are weight parameters to be optimized, and y represents the final output.

3.3. Generative adversarial networks

The GAN model is recognized by the capability of generating diverse prediction results. In the framework of a generic GAN model, two neural networks (generator and discriminator) are composed and they are trained in an adversarial manner. To ensure the diversity of predictions, the generator is fed with data randomly sampled from a prior distribution $p(z)$, usually a multivariate Gaussian distribution. Next, the generator outputs multiple prediction results to be distinguished from real observations by the discriminator. Therefore, the generator is trained to produce predictions that are as realistic as possible for successfully "fooling" the discriminator, while the discriminator is trained to distinguish the predictions of the generator as accurately as possible. The model training can be formulated as an minimax optimization problem as expressed in (4) and (5):

$$\min_{\theta}\max_{\varphi} V(\theta, \varphi) \tag{4}$$

$$V(\theta,\varphi) = -\left(\mathbb{E}[D_\varphi(y)] - \mathbb{E}[D_\varphi(G_\theta(z))]\right)(5)G_\theta(z))]\big)(5)\mathbb{E}[D_\varphi(y)] - \mathbb{E}[D_\varphi$$

where φ and θ are the network parameters of the discriminator D_φ and the generator G_θ, respectively, and $V(\theta,\varphi)$ denotes the objective function that needs to be optimized and z is the random noise.

4. GAPGAN for short-term predictive wind scenario generation

To generate accurate and diverse wind scenario predictions, we propose the GAPGAN model as shown in Fig. 4, which consists of three modules. In the first module, data processing is conducted to establish the encoding of temporal information, generate random noises and split the dataset. Next, in the SG model construction module, the generator network composed of a temporal convolution (TC) model and a GC model is constructed to effectively mine temporal and spatial relationships of the wind field, respectively. Finally, the established SG model is trained using a GAN framework, combined with the proposed adaptive training mechanism and predictive modeling capability. Details of the three modules involved are next explained.

4.1. Data processing module

The data used in the model are collected from the SCADA system of wind turbines, which records the historical wind speed at multiple locations. To perform the predictive modeling, the sliding window method

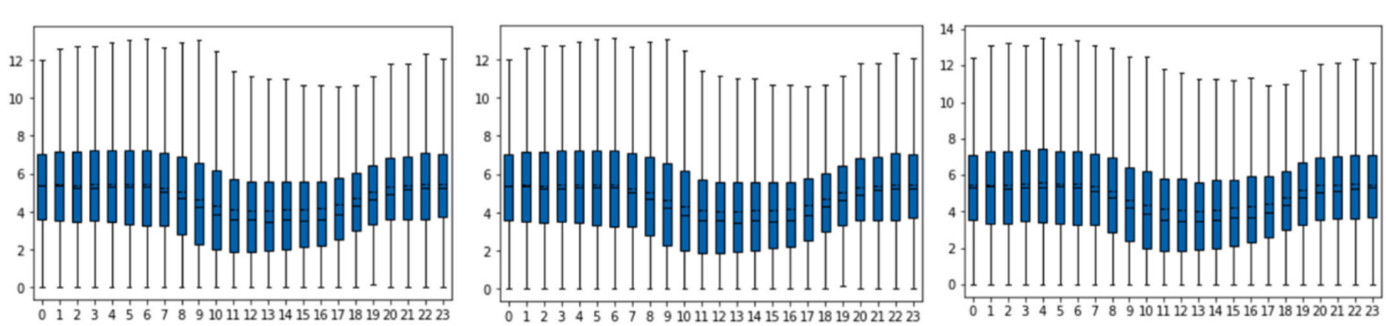


Fig. 5. Wind speed distributions of three wind turbines at different time of day.

Fig. 6. Dataset partitioning.

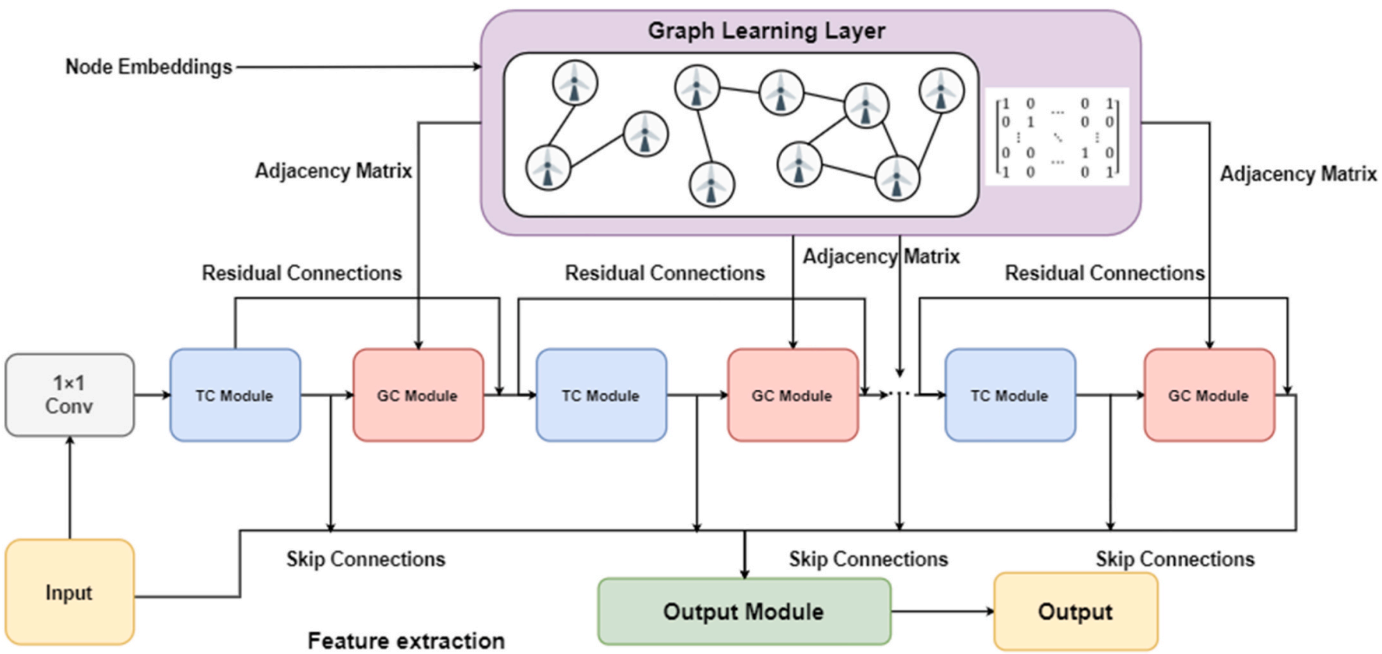


Fig. 7. The illustration of the scenario generation model network architecture.

is first used to split the standardized wind speed data and obtain a set of input-output data pairs for model training. The historical wind speed data is denoted by $Hw = \{hw1, hw2, \ldots, hwK\}$, where $h_{wi} \in \mathbb{R}^{N\times lh}$ denotes the wind speed data at location i, K denotes the total number of locations, and N and lh represent the number and the length of input data sequences, respectively. The future wind speed observations of each input sequences are regarded as the prediction target and are denoted by $P = \{p1, p2, \ldots, pK\}$, where $p_i \in \mathbb{R}^{N\times lp}$ represents the short-term future wind speed at location i accordingly, and lp denotes the prediction horizon.

In addition to the wind speed time series data, the information of time is also considered as the model input. Based on the preliminary data analysis as shown in Fig. 5, the 24-h wind speed distributions show similar patterns at different locations. Thus, we encode the time of day by scaling the time from 00:00:00 to 23:59:59 into the range [0, 1] for simplicity and denote the time encoding by H_{time}. The input data of the model is finally obtained by aggregating H_{time} and H_w, which makes $H=[H_{time}, H_w]$.

Finally, the original dataset is divided into a training set $\{D_{train}, P_{train}\}$, a validation set $\{D_{val}, P_{val}\}$, and a test set $\{D_{test}, P_{test}\}$ according to a pre-set proportion, which is considered as 70%:10%:20%. The detail of the dataset partitioning is shown in Fig. 6.

4.2. 2 scenario generation model construction

To establish an accurate SG model, exploiting spatiotemporal relationships requires appropriate data structure and model architecture accordingly. In this module, we refer to the multivariate graph network-based model (MTGNN) [34] as the backbone of the SG model, which can later be understood as the generator in a GAN framework.

The SG model architecture is generally composed of a graph learning layer, several TC modules and GC modules, as shown in Fig. 7. The input data are first fed into a convolutional layer, and next into several blocks sequentially, each of which is composed of a TC module and a GC module. Residual connections are applied between the blocks for smoother gradient backpropagations. An additional graph learning layer is constructed to extract relationships between the graphs in different layers, which are used to represent the adjacency matrix of wind turbines. Details of the modules involved are introduced next.

4.2.1. Graph learning layer

The graph learning layer is applied to adaptively learn to construct a graph adjacency matrix for the wind field data, which facilitates to further extract the hidden spatial relationships between wind speed time series at different locations. The adjacency matrix describes which nodes should be connected in a graph. In a wind speed prediction problem, we only consider undirected relationships in the graph, and the adjacency matrix learning process [34] is expressed as (6)–(10):

$$M_1 = tanh\,(\alpha E_1 \Theta_1) \tag{6}$$

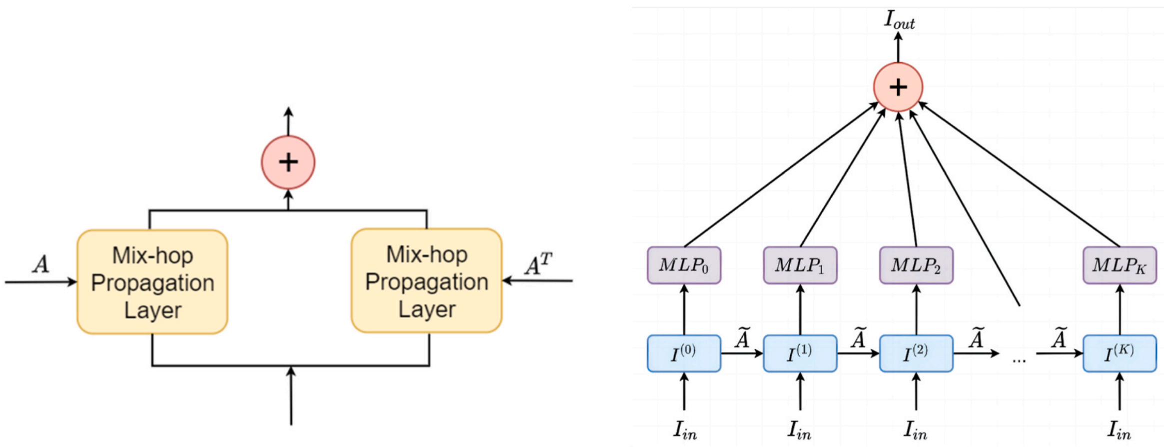
$$M_2 = tanh\,(\alpha E_2 \Theta_2) \tag{7}$$

$$A = ReLU\left(tanh\left(\alpha\left(M_1 M_2^T - M_2 M_1^T\right)\right)\right) \tag{8}$$

$$idx = argtopk\,(A[i,:]), \text{for } i = 1, 2, \cdots, N \tag{9}$$

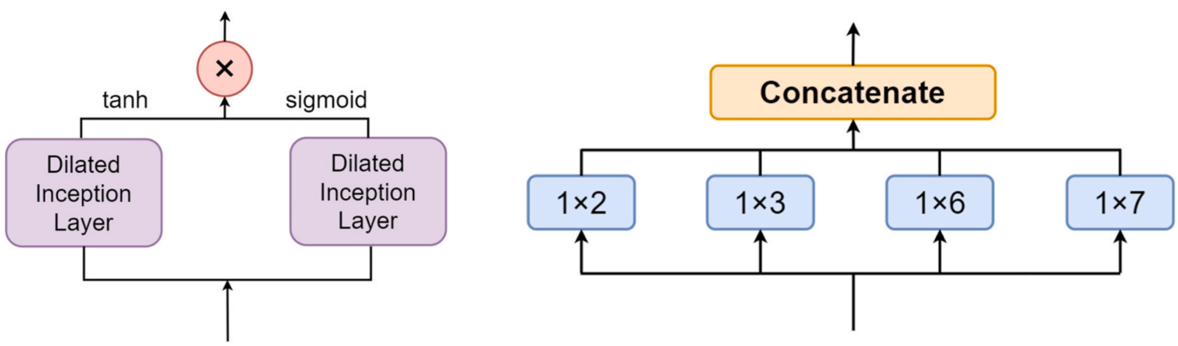
$$A[i, -idx] = 0 \tag{10}$$

where E_1 and E_2 represent the randomly initialized node embeddings, Θ_1 and Θ_2 are model parameters to be optimized for extracting features from node embeddings, α is a hyper-parameter for controlling the



(a) GC module (b) Mix-hop propagation layer

Fig. 8. The illustration of GC module and mix-hop propagation layer.



(a) TC module (b) Dilated inception layer

Fig. 9. The illustration of (a) TC module and (b) dilated inception layer.

saturation rate of the activation function, N is the total number of wind turbines and the random noise dimension, representing the maximum number of input nodes, and $argtopk(\cdot)$ returns indices of the top-k largest values of a vector denoted by idx where k is a preset hyperparameter. Eq. (10) means that the remaining values not indexed by idx in the adjacency matrix A are all set to 0. The resulting adjacency matrix A is next used in GC modules.

4.2.2. Graph convolution module

The GC module aims to fuse the information between adjacent nodes and learn the spatial relationships of the wind field data. The GC operation is conducted by combining the output of two mix-hop propagation layers, as shown in Fig. 8(a). More specifically, as shown in Fig. 8(b), each mix-hop propagation layer performs two operations: information dissemination and information selection. The information dissemination operation [34,35] is expressed as (11):

$$I^{(i)} = \beta I_{in} + (1-\beta)\widetilde{A} I^{(i-1)} \tag{11}$$

where β is the ratio of maintaining the original state, I_{in} is the state information of the initial input, $I^{(i)}$ is the hidden state of data of layer i, $\widetilde{A}$ represents the adjacency matrix, which is either A or A^T. Next, the information selection operation [34,35] is expressed in (12):

$$I_{out} = \sum_{i=0}^{m} I^{(i)} W^{(i)} \tag{12}$$

where m is the depth of propagation, $W^{(i)}$ is model parameters of layer i to be optimized, and I_{out} is the hidden state of the output.

4.2.3. Temporal convolution module

Unlike conventional convolutional layers that apply 2-D kernels to process image-like 2-D or 3-D tensors, the TC module uses 1-D convolution kernels to extract temporal features of multivariate time series. The TC module consists of two dilated inception layers, one of which is followed by a tangent hyperbolic function and the other by a sigmoid function. More specifically, as shown in Fig. 9 (b), in each dilated inception layer, four convolutional operations using kernels of various shapes are conducted and their results are concatenated. Convolutional operations of dilated inception layers are expressed in (13).

$$p_{out}(j) = p_{in} * f_{1\times n} = \sum_{r=0}^{b-1} f_{1\times n}(r) p_{in}(j - dil \times r) \tag{13}$$

where p_{in} and p_{out} are the input and output sequence data, respectively, $p_{in}(j)$ and $p_{out}(j)$ separately represent the j-th element in p_{in} and p_{out}, $fk_{1\times n}$ is a 1-D convolution kernel of length b, and dil represents the expansion factor. It is worth noting that since the four convolutional operations should lead to outputs of different dimensions, to make the summation feasible, output values of all four convolutional operations are truncated and connected according to the smallest dimension of output data.

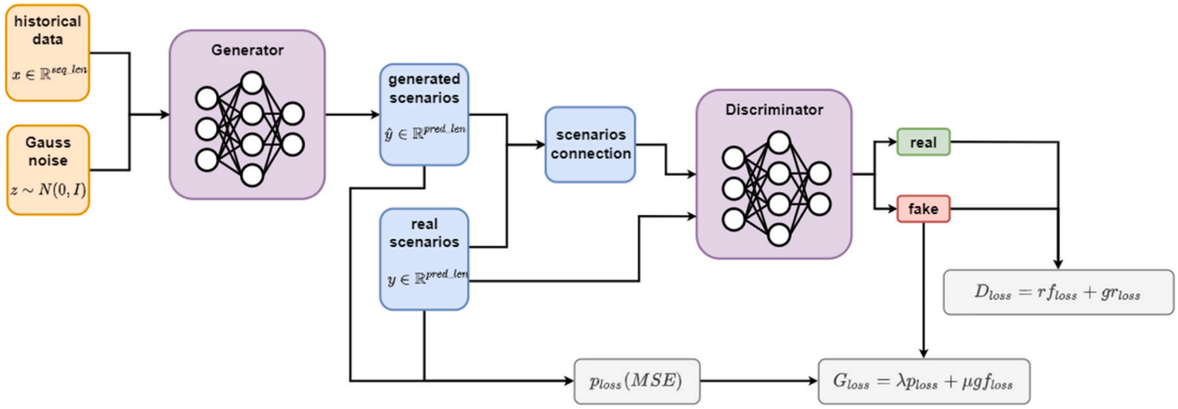


Fig. 10. The framework of predictive generative adversarial network training process.

4.3. Adaptive predictive generative adversarial network training

After processing the input data and constructing a SG model, the SG model are next trained via an adaptive predictive GAN (APGAN) training process, which adopts three mechanisms: PGAN, adaptive model training, and optimal model determination. The proposed three mechanisms are next explained in detail.

4.3.1. Predictive generative adversarial network training

Despite the fact that conventional GANs can be used to model the joint distribution of multiple input variables, they are not applicable for performing predictive modeling, which requires the generation of different outputs conditional on a given set of continuous historical observations. Also, although conditional CGANs, an advanced variant of GANs, can generate samples based on different input conditions, these conditions are usually formulated as discrete class labels rather than continuous values such as wind speed, thus not applicable to the predictive SG of wind speed, either.

Particularly, in a predictive modeling approach, a SG model should produce wind speed sequences that are very close to short-term future real values while ensuring the scenario set could present the prediction uncertainty accurately. To address this problem and overcome limitations of GAN and CGAN models, we propose a PGAN framework in this paper. On the basis of a conventional GAN, an additional prediction error term is added to the loss function of the generator to yield the coordination between the generative ability and prediction accuracy, which is the most significant feature of PGAN.

The proposed PGAN framework is illustrated in Fig. 10. As shown in Fig. 10, the generator model is applied to predict future wind scenarios, and the discriminator is trained to accurately distinguish between real observations and scenario predictions. The input to the generator includes the historical wind speed data from multiple locations and noise data denoted by z, which are randomly generated from multivariate normal distribution to ensure diversity of scenarios. On the other side, scenario predictions and real observations are fed into the discriminator to determine if the predictions look authentic.

Generating accurate wind speed scenario predictions is equivalent to making the generated scenarios $\hat{y}$ indistinguishable from the real future observations as judged by a highly judicious discriminator. To achieve this goal, we propose the loss functions of the discriminator D_{loss} and the generator G_{loss}, as expressed in (14)–(16).

$$\hat{y} = G_\theta(x, z) \tag{14}$$

$$D_{loss} = \mathbb{E}_{y\sim P_r}[D_\varphi(y)] + \mathbb{E}_{\hat{y}\sim P_f}[1 - D_\varphi(\hat{y}, y)] \tag{15}$$

$$G_{loss} = \lambda(\hat{y} - y)^2 + \mu\mathbb{E}_{\hat{y}\sim P_f}[D_\varphi(\hat{y}, y)] \tag{16}$$

where x is historical wind speed data, z is random noise data, G_θ is the generator network with parameter θ, $\hat{y}$ is the predicted scenarios, y is the real future wind speed data, D_φ is the discriminator with parameter φ, $\mathbb{E}_{y\sim P_r}[D_\varphi(y)]$ is the loss of judging the real data as fake, $\mathbb{E}_{\hat{y}\sim P_f}[1 - D_\varphi(\hat{y}, y)]$ is the loss of wrongly identifying the predicted scenarios as real, $\mathbb{E}_{\hat{y}\sim P_f}[D_\varphi(\hat{y}, y)]$ is the loss of correctly distinguishing the predicted scenarios, and λ and μ are hyperparameters.

4.3.2. Adaptive training mechanism

As indicated by (15),(16), the generator aims to minimize the $\mathbb{E}_{\hat{y}\sim P_f}[D_\varphi(\hat{y}, y)]$ while the discriminator aims to maximize it. Since PGAN training is essentially a zero-sum non-cooperative game, the training of the generator and discriminator is very difficult to converge, and their losses tend to fluctuate heavily.

In a generic GAN framework, it is found that the discriminator is very likely to dominate the training process, which makes it easy to identify all scenarios predicted by the generator as false. Thus, the generator is unable to find an optimization direction. To stabilize the training process and keep a balance between the generator and discriminator, we propose an adaptive training g mechanism by controlling the training round of the discriminator.

The proposed adaptive training method, i.e., APGAN training, is described in Algorithm 1 and displayed in Appendix B. In the APGAN training process, the generator and discriminator are optimized iteratively, and their loss values are monitored. The proposed adaptive training strategy specifies that if the loss function of the generator is continuously increased for a certain number of epochs, say u_1, the discriminator training will be temporally stopped. After several rounds of the generator training, the training of the discriminator will be resumed.

4.3.3. Optimal model selection

Since model performance inevitably fluctuates during model training, the optimal SG model may not be the final model at the time of training convergence, but may arise during the training process and should be thus selected. However, due to the multiple uses of SG prediction results, it is difficult to evaluate the quality of models based on a single criterion. Furthermore, existing SG evaluation indices are usually used to assess generic 24-h wind scenarios, while the predictive SG model lacks a unified evaluation index. Considering that the predicted scenarios should both precisely match the future wind speed results and reflect the prediction uncertainty, we propose an evaluation function for SG models, as expressed in (17):

$$E = \frac{1}{N_{val}}\sum_{i=1}^{N_{val}} \alpha_1 \bullet RMSE\left(P_{val}{}^{(i)}, \widehat{P}_{mean}{}^{(i)}\right) + \alpha_2 \bullet DTW\left(P_{val}{}^{(i)}, \widehat{P}_{mean}{}^{(i)}\right)$$

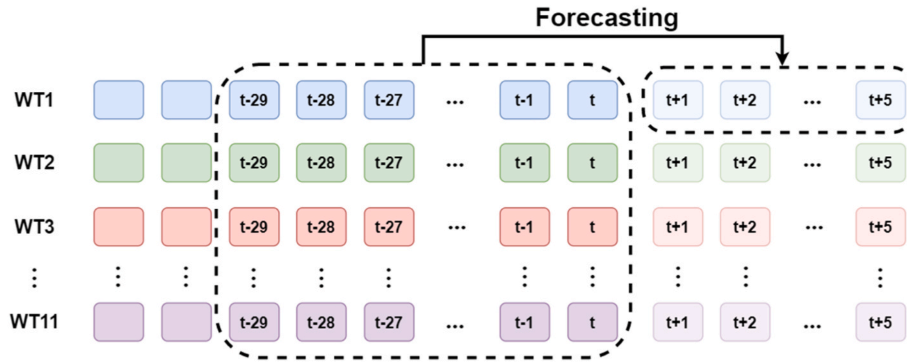


Fig. 11. The illustration of the predictive scenario generation scheme.

$$+\alpha_3 \bullet MSCWC\left(P_{val}{}^{(i)}, \widehat{P}_{val}{}^{(i)}\right) \tag{17}$$

where N_{val} is the size of validation set, $\widehat{P}_{mean}{}^{(i)}$ and $\widehat{P}_{val}{}^{(i)}$ represent the mean of the predicted scenarios and the set of predicted scenarios for the i-th input data, respectively, $RMSE(\bullet,\bullet)$ computes the root mean square error (RMSE) between two series of data, $DTW(\bullet,\bullet)$ computes the dynamic time warping (DTW) between two time series for assessing the shape similarity, $MSCWC(\bullet,\bullet)$ refers to the multi-step coverage width-based (MSCWC) metric for evaluating the interval prediction performance of the generated scenarios, and $\alpha_1, \alpha_2, \alpha_3$ are preset hyperparameters. Formulae of $RMSE, DTW, MSCWC$ are described in detail in Section 5.2.

After carefully assessing the variations in magnitudes among RMSE, MSCWC, and DTW, our preliminary experimental trials suggest assigning weights of 1, 1, and 0.2 to them, respectively. Nevertheless, all hyperparameters are flexible and can be adjusted based on their relative importance. If a higher prediction accuracy is desired, the weight for RMSE can be increased. If superior interval prediction performance is the goal, a higher weight should be given to the MSCWC. Similarly, the weight of DTW can be increased if shape similarity is prioritized higher.

Finally, the optimal model selection process is described in Algorithm 2, which is displayed in Appendix C.

5. Computational studies

In this section, experimental studies are conducted based on real data collected from a wind farm to validate the effectiveness of the proposed GAPGAN-based SG method. The data used in the experiment, experimental settings, and detailed discussions of the results are next presented.

5.1. Dataset description

The raw data used in this paper were collected from the SCADA system of 11 wind turbines in a wind farm in the United States for the period from January 26, 2016 and June 5, 2017. The original wind speed data were recorded at 8-s intervals. To denoise the data and to facilitate up to 30 min-ahead short-term wind speed predictions, we constructed a new wind speed dataset with 6-min intervals by taking a mean value of the raw wind speed data every 6 min, which forms the basis for all experiments conducted in this study. Finally, the dataset is divided into a training set, a validation set and a test set according to the ratio of 70%:10%:20% as explained in Section 4.1.

5.2. Experimental settings

In this section, the experimental setup regarding the benchmarking models, prediction task, and error metrics are introduced. Details of the hyperparameter settings for the proposed GAPGAN model are provided in Appendix A.

Benchmarking models: five benchmarking wind speed prediction models are implemented for comparisons, which include two classical models: variational autoencoders (VAE) [36] and naïve prediction, and three state-of-the-art (SOTA) models: CNN-GAN [3], TCN-GAN [31], and CNN-LSTM-GAN [37]. The considered three SOTA SG models are similarly developed using the GAN framework; however, their generators are constructed using CNN, TCN, and CNN-LSTM networks, respectively. Hyperparameters of all models are carefully determined via trial-and-errors.

Prediction task: to conduct predictive SG for wind speed, we employ the most recent wind speed data collected from all sites over the preceding 3 h as the model input. The input data is structured as a multivariate time series, with each series comprising 30 data points and featuring a resolution of 6 min as described in Section 5.1. Utilizing this information, we execute multistep predictions for time horizons of 6 min, 12 min, 18 min, 24 min, and 30 min ahead. We consider to generate 50 short-term wind speed prediction scenarios for each wind speed series to be predicted in our experiment, and the number of scenarios can be varied in practice. The prediction scheme is illustrated in Fig. 11.

Evaluation metrics: To comprehensively analyze the effectiveness and utility of the proposed GAPGAN-based SG method, we apply six evaluation metrics, namely, mean absolute error (MAE), root mean square error (RMSE), dynamic time warping (DTW), multi-step coverage probability (MSCP), prediction interval normalized averaged width (PINAW) and multi-step coverage width-based criterion (MSCWC). Among these metrics, MAE and RMSE, which are widely applied to assess the point prediction error as expressed in (18),(19), are used to evaluate if the overall predicted trend is accurate.

$$MAE = \frac{1}{l}\sum_{i=1}^{l}\left|y_i - \frac{1}{t}\sum_{j=0}^{t}\widehat{y}_i^{(j)}\right| \tag{18}$$

$$RMSE = \sqrt{\frac{1}{l}\sum_{i=1}^{l}\left(y_i - \frac{1}{t}\sum_{j=1}^{t}\widehat{y}_i^{(j)}\right)^2} \tag{19}$$

where y_i is the i-th step real wind speed values in future, $\widehat{y}_i^{(j)}$ is the i-th step predicted value of the j-th average scenario in future, l is the length of the prediction horizon, and t is the number of generated scenarios.

MSCP and PINAW are a pair of complimentary indicators to evaluate the interval prediction result from different perspectives. MSCP is used to assess the reliability of the prediction interval, measuring the extend to which observations fall within the prediction intervals. However, this aspect prefers sufficiently wide prediction intervals to produce advantageous MSCPs, but of little practical significance due to the low confidence. On the other side, PINAW is used to evaluate the sharpness of the interval prediction, that is, PINAW prefers narrow predicted intervals. To strike a balance between the reliability and sharpness of interval predictions, MSCWC is also introduced to evaluate interval prediction

Table 2
MAE for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	0.246	0.231	0.230	**0.250**	0.228	0.250	0.211	0.211	0.198	0.194	0.200	0.223
CNN-LSTM-GAN	0.246	0.226	0.230	0.254	0.227	0.252	0.209	0.211	0.200	0.198	0.202	0.223
TCN-GAN	0.249	0.227	0.231	0.254	0.229	0.252	**0.202**	0.213	0.199	0.199	0.208	0.224
GAPGAN	**0.245**	**0.225**	**0.229**	0.253	**0.225**	**0.249**	0.210	**0.203**	0.198	**0.190**	0.198	**0.220**
VAE	0.256	0.231	0.233	0.255	0.229	0.257	0.211	0.218	0.204	0.205	0.207	0.228
NAIVE	0.252	0.233	0.234	0.255	0.229	0.259	**0.202**	0.206	**0.196**	0.197	**0.196**	0.224

Table 3
RMSE for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	0.284	0.264	0.264	**0.286**	0.260	0.288	0.243	0.240	**0.225**	0.222	0.228	0.255
CNN-LSTM-GAN	**0.283**	**0.259**	0.263	0.291	0.259	0.289	0.239	0.239	0.227	0.227	0.230	0.255
TCN-GAN	0.284	0.261	0.265	0.294	**0.254**	0.290	0.235	0.241	0.226	0.230	0.231	0.256
GAPGAN	**0.283**	**0.259**	**0.261**	0.290	0.258	**0.286**	0.240	**0.235**	0.228	**0.218**	0.226	**0.253**
VAE	0.289	0.264	0.264	0.293	0.264	0.291	0.238	0.246	0.231	0.228	0.233	0.258
NAIVE	0.286	0.265	0.264	0.291	0.260	0.295	**0.231**	0.245	0.233	0.233	**0.224**	0.257

comprehensively. The MSCP, PINAW, and MSCWC are expressed in (20)–(23)

$$MSCP = \frac{1}{l}\sum_{i=1}^{l}\sum_{j=1}^{t} c_i^{(j)} \tag{20}$$

$$PINAW = \frac{1}{lR}\sum_{i=1}^{l}|U_i - L_i| \tag{21}$$

where $c_i^{(j)} = 1$ when $\hat{y}_i^{(j)} \in [L_i, U_i]$, otherwise $c_i^{(j)} = 0$, U_i and L_i are the upper and lower bounds of the i-step interval prediction in future, l is the prediction horizon, and R is a constant.

$$MSCWC = PINAW\left(1 + \gamma e^{-\eta(MSCP-\mu)}\right) \tag{22}$$

$$\gamma = \begin{cases} 0 \; MSCP \geq \mu \\ 1 \; MSCP < \mu \end{cases} \tag{23}$$

where μ is the level of confidence, and γ and η are hyperparameters affecting the degree of punishment. Next, DTW evaluates the extend to which the predicted wind speed trend follows the shape of real observations, as expressed in (24)–(25)

$$\gamma(q_v, c_u) = d(q_v, c_u) + min\{\gamma(q_{v-1}, c_{u-1}), \gamma(q_{v-1}, c_u), \gamma(q_v, c_{u-1})\} \tag{24}$$

$$DTW = \gamma(q_m, c_n) \tag{25}$$

where q_i, i=1, …, m and c_i, i=1, …, n are the elements in time series Q and C, respectively, $\gamma(q_v, c_u)$ is the cumulative distance from (q_1, c_1) to (q_v, c_u), and $d(q_v, c_u)$ is the difference between q_v and c_u.

5.3. Experimental results

In this section, we first examine the prediction performance of all considered models in terms of prediction accuracy, uncertainty, diversity, and shape similarity to validate the effectiveness of the proposed GAPGAN-based SG model in Sections 5.3.1 - 5.3.3, respectively. Next, an ablation study comparing APGAN and PGAN is conducted in Section 5.3.4 to verify the advantages of the proposed adaptive training mechanism. Discussions and conclusions are provided in corresponding sections.

5.3.1. Point prediction accuracy

Since accuracy is the primary concern for the wind speed SG prediction, we first use the point prediction evaluation metrics, MAE and RMSE, to evaluate the overall prediction accuracy by comparing the

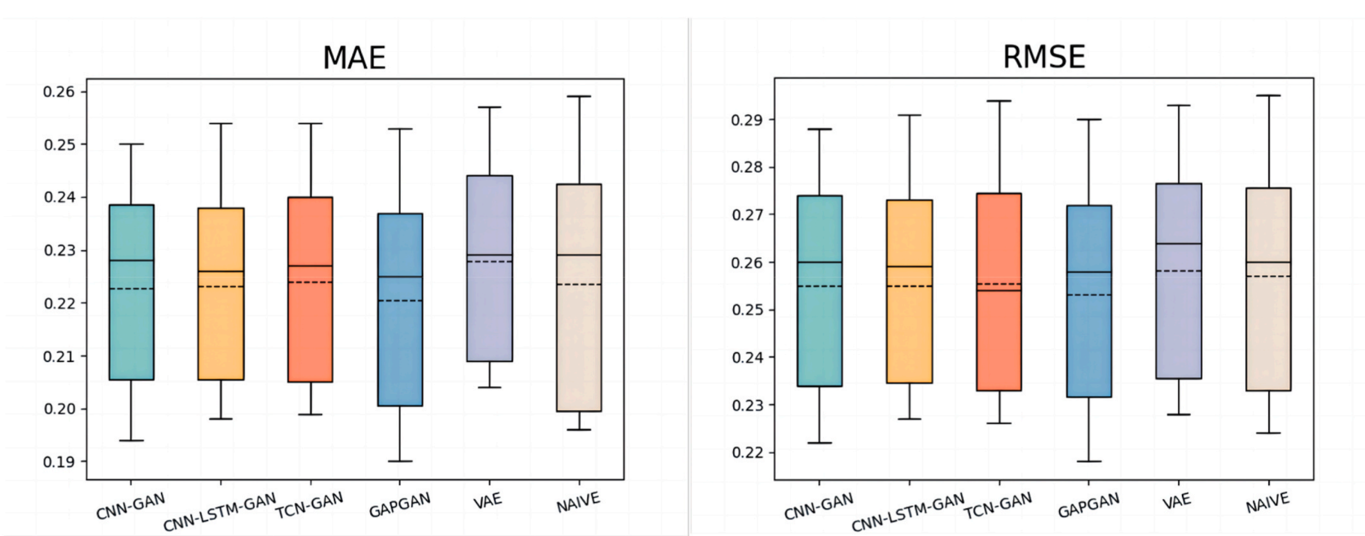


Fig. 12. Box-whisker plots of MAE and RMSE distributions for 11 wind turbines.

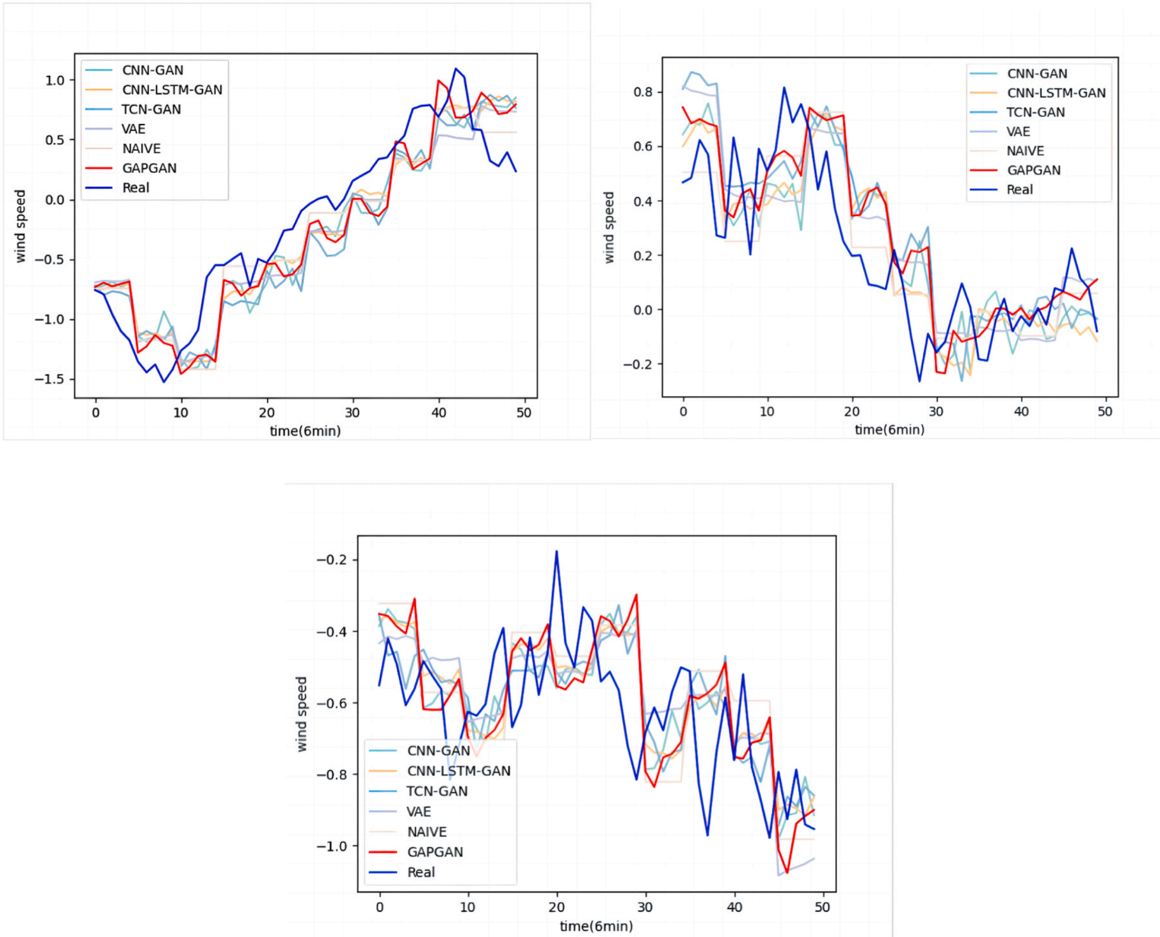


Fig. 13. 30-min ahead wind speed predictions for three different periods of wind speed.

average scenario with actual observations. Tables 2–3 list results of MAEs and RMSEs for 11 wind turbines on the test set, as well as their mean values for all wind turbines, where the best results are underlined and bold.

As shown in Tables 2–3, the proposed GAPGAN method yields the lowest average MAE and RMSE for 11 wind turbines, suggesting its highest prediction accuracy overall. However, GAPGAN does not achieve optimal results on all wind turbines, such as wind turbines 4, 7, 9, and 11. Moreover, among the four GAN-based models, GAPGAN shows advantages on the largest number of wind turbines, indicating its generality. In contrast, TCN-GAN using TCN alone without combining GNN or CNN modules could not present satisfactory accuracy, except for the optimal performance achieved at wind turbine 7, which manifests that spatial features are not sufficiently exploited. Despite the fact that the CNN module could extract spatial relationships of the wind field in a different way and could occasionally obtain lower RMSEs for some wind turbines, the CNN module is still inferior to GNN-TCN used in GAPGAN overall. In addition, as can be seen in Table 2, the naïve prediction could occasionally lead to accurate average predictions on several wind turbines, yet it has the second largest average MAE and RMSE, indicating unstable performance and lack of generality. Finally, the highest MAE and RMSE yielded by VAE demonstrate that GAN is a more effective generative framework for predictive SGs for wind speed.

To better compare the generality of considered models, distributions of RMSEs and MAEs for 11 wind turbines, box-whisker plots are presented in Fig. 12. As shown in Fig. 12, GAPGAN yields the lowest box among all models, proving its generality for different wind turbines.

In Fig. 13, the results of 30-min ahead point predictions for three randomly selected periods of wind speed are presented, which show that the average trend of the wind speed predictions yielded by GAPGAN could well reflect the actual wind speed trend.

5.3.2. *Uncertainty evaluation accuracy and diversity*

In this section, the uncertainty evaluation accuracy and diversity of short-term predictive SG results are presented and analyzed. First, since SG can be used to assess the future uncertainty of wind speed, we regard quantile values of the scenario prediction set as interval prediction results. Tables 4–6 list the MSCP, PINAW, and MSCWC result for all

Table 4
MSCP for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	**0.602**	**0.675**	**0.588**	**0.560**	**0.664**	**0.542**	**0.618**	**0.590**	**0.721**	**0.615**	0.639	**0.620**
CNN-LSTM-GAN	0.432	0.454	0.510	0.455	0.470	0.452	0.522	0.492	0.547	0.491	0.507	0.485
TCN-GAN	0.415	0.505	0.555	0.483	0.423	0.525	0.535	0.517	0.568	0.493	0.548	0.506
GAPGAN	0.425	0.519	0.560	0.489	0.444	0.507	0.572	0.563	0.580	0.488	**0.644**	0.526
VAE	0.384	0.456	0.408	0.406	0.391	0.387	0.494	0.434	0.419	0.447	0.443	0.424
NAIVE	–	–	–	–	–	–	–	–	–	–	–	–

Table 5
PINAW for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	0.101	0.112	0.092	0.089	0.104	0.085	0.087	0.084	0.104	0.081	0.089	0.093
CNN-LSTM-GAN	0.065	**0.066**	0.073	0.070	0.065	0.069	0.068	0.065	0.071	0.062	0.066	0.067
TCN-GAN	0.077	0.069	0.076	0.074	0.065	0.073	0.071	0.068	0.075	0.059	0.079	0.072
GAPGAN	**0.053**	0.074	0.081	0.074	**0.055**	0.077	0.076	0.071	0.076	0.056	0.086	0.071
VAE	0.059	0.069	**0.061**	**0.065**	**0.055**	**0.060**	**0.064**	**0.057**	**0.053**	**0.056**	**0.057**	**0.060**
NAIVE	–	–	–	–	–	–	–	–	–	–	–	–

Table 6
MSCWC for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	**0.270**	0.236	0.252	**0.252**	0.239	**0.261**	0.222	0.225	**0.201**	**0.197**	0.210	**0.233**
CNN-LSTM-GAN	0.273	0.253	0.229	0.278	0.261	0.278	0.211	**0.203**	0.208	0.215	0.208	0.238
TCN-GAN	0.282	0.268	0.230	0.277	**0.235**	0.275	0.213	0.213	0.222	0.218	0.206	0.240
GAPGAN	0.281	**0.232**	**0.231**	0.268	0.241	0.262	0.217	0.212	0.218	0.202	0.202	**0.233**
VAE	0.276	0.247	0.233	0.280	0.265	0.274	**0.210**	0.207	0.218	0.209	**0.197**	0.238
NAIVE	–	–	–	–	–	–	–	–	–	–	–	–

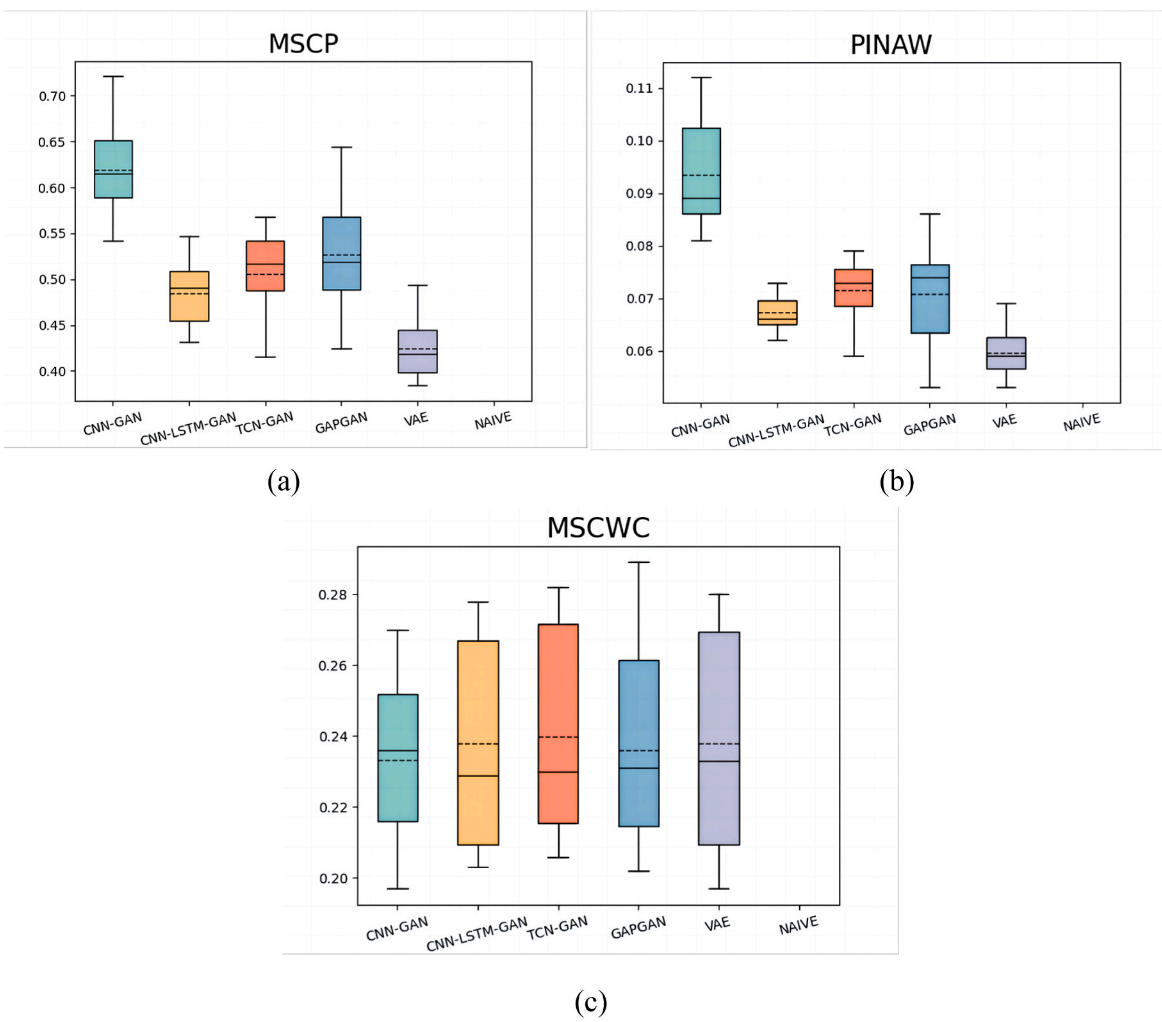
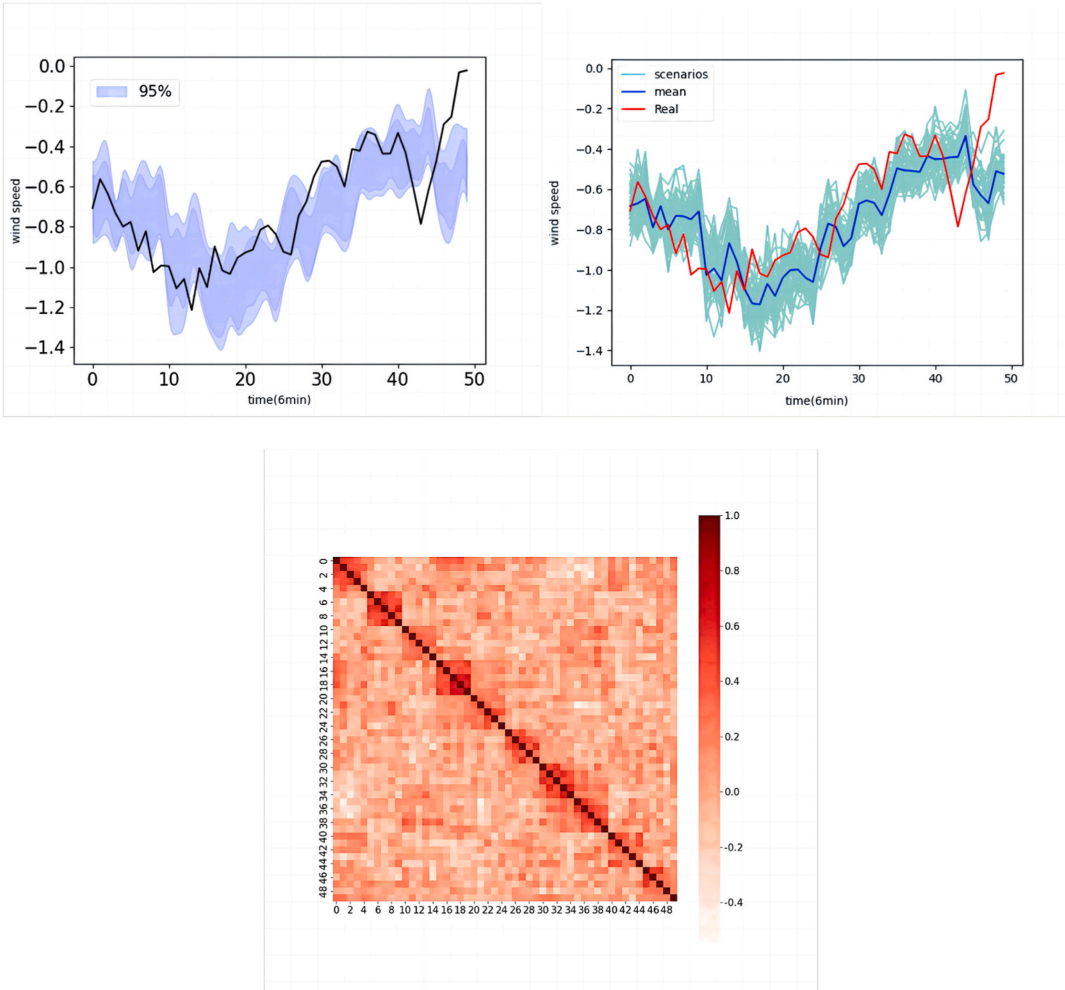


Fig. 14. Box-whisker diagrams of (a) MSCP, (b) PINAW and (c) MSCWC distributions.

considered models using the test set except for results of the naïve prediction method as it can only provide a single deterministic prediction result.

Firstly, the results in Table 4 clearly show that CNN-GAN presents the highest MSCP value of 62.0%, indicating that its scenario range could cover the most real observations for the largest number of wind turbines. However, the largest PINAW value of 0.093 given by CNN-GAN shown in Table 5 manifests the price, i.e., the scenario range becomes accordingly large, which undermines the sharpness of the prediction interval. In comparison, GAPGAN obtains the second best MSCP



(a) CNN-GAN

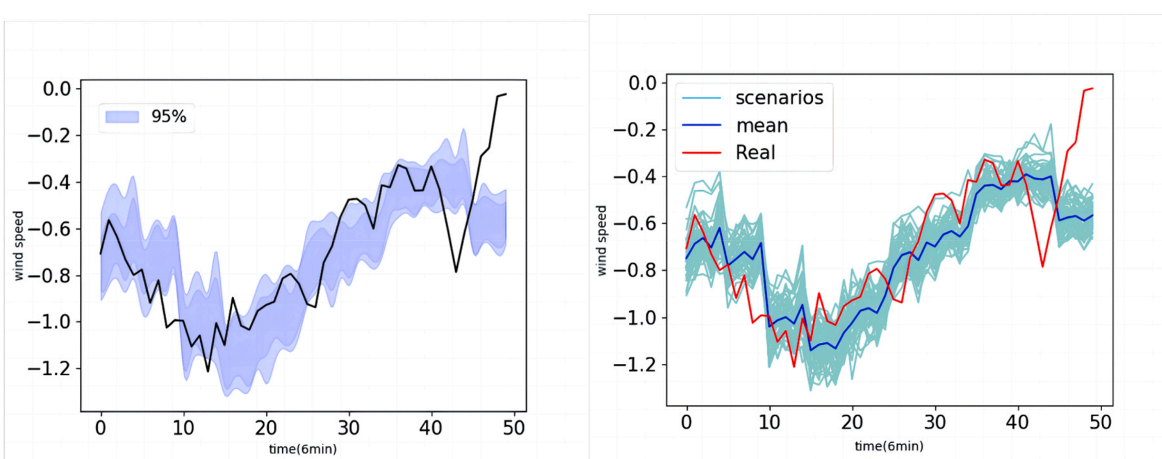
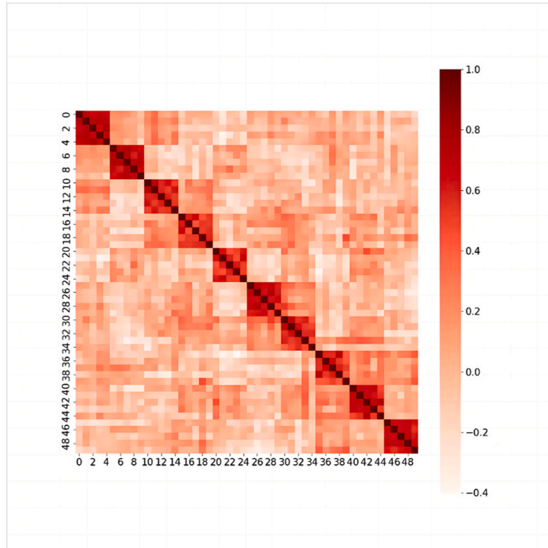
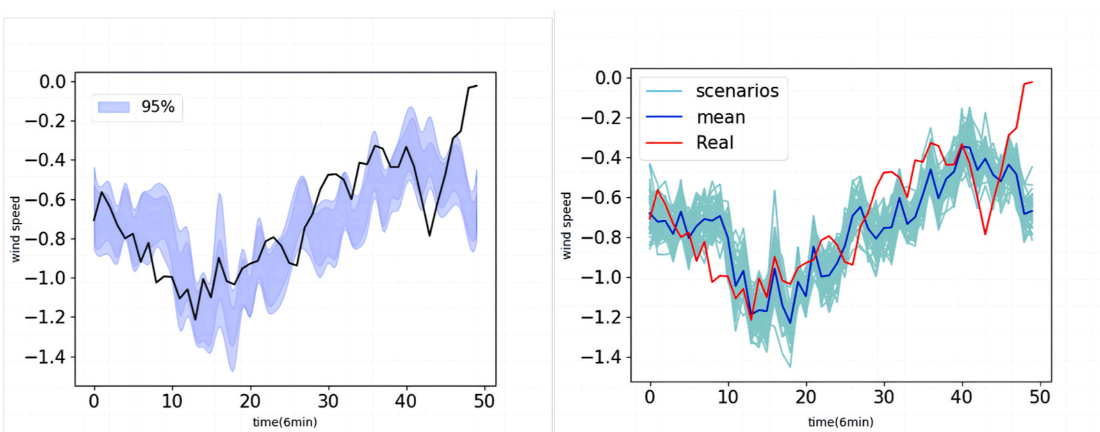


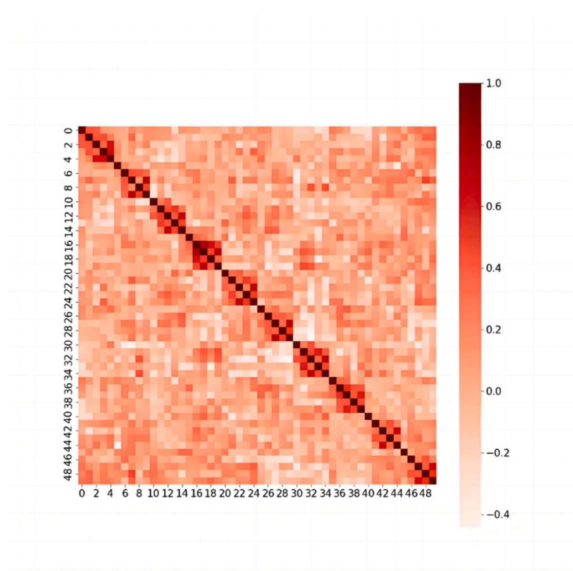
Fig. 15. The 30-min ahead wind speed interval predictions, scenario generation predictions, and correlation heat maps for generated scenarios for five considered models.

value and the third best PINAW value, and does not seem to have an absolute advantage in either of aspects. However, as a result, despite the fact that both CNN-GAN and GAPGAN achieve the lowest MSCWC value of 0.233, CNN-GAN neglects the sharpness aspect while GAPGAN seeks a balance between reliability and sharpness. In addition, the scenario range predicted by VAE is the narrowest, shown by its smallest PINAW value; meanwhile, the reliability is severely damaged and the MSCP is the lowest, only reaching 0.424. Besides, CNN-LSTM-GAN and TCN-GAN obtain similar interval prediction widths as the proposed GAP-GAN, with the PINAW values of around 0.70 shown in Table 5; however,



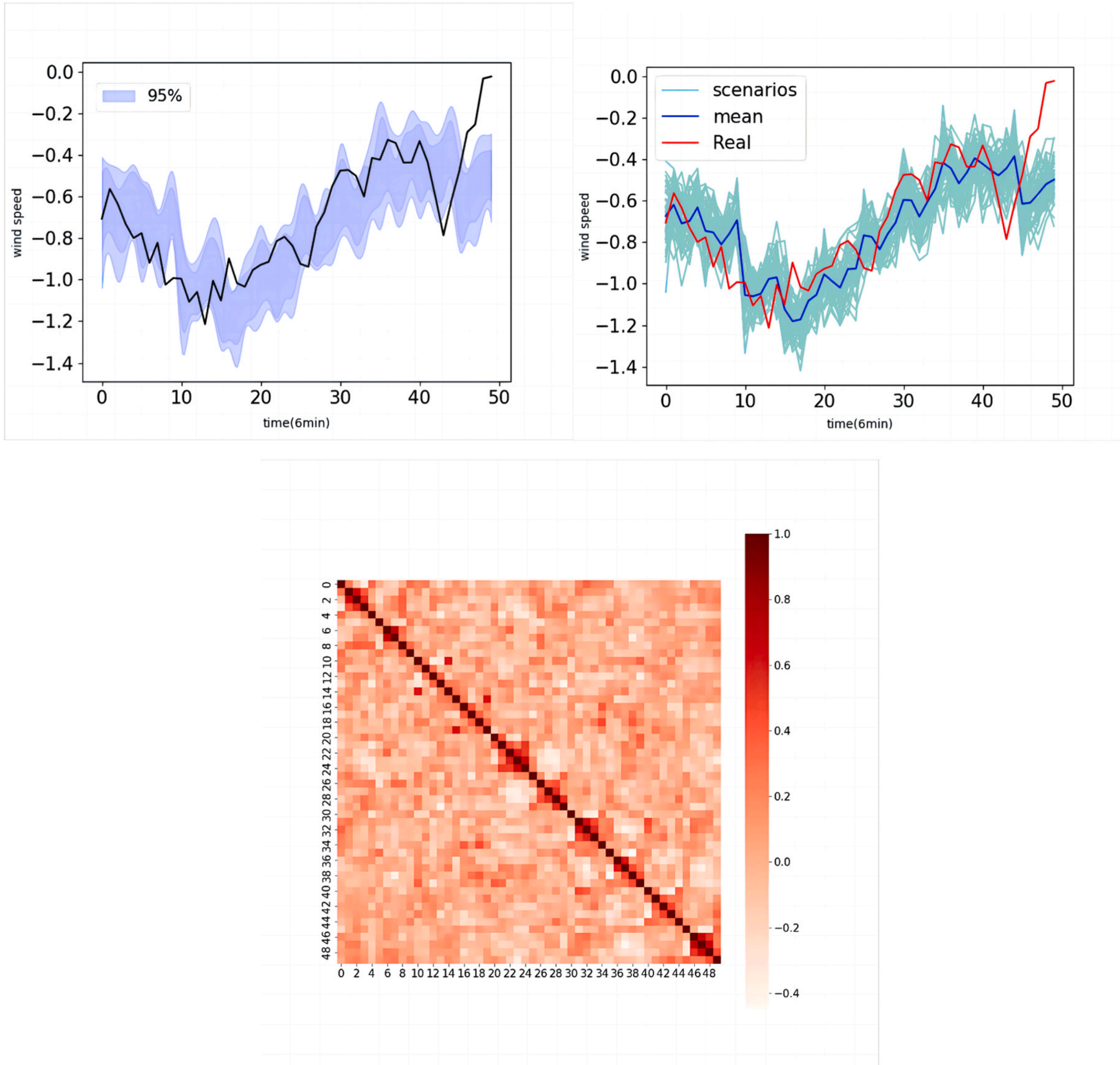
(b) CNN-LSTM-GAN





(c) TCN-GAN

Fig. 15. (*continued*).



(d) GAPGAN

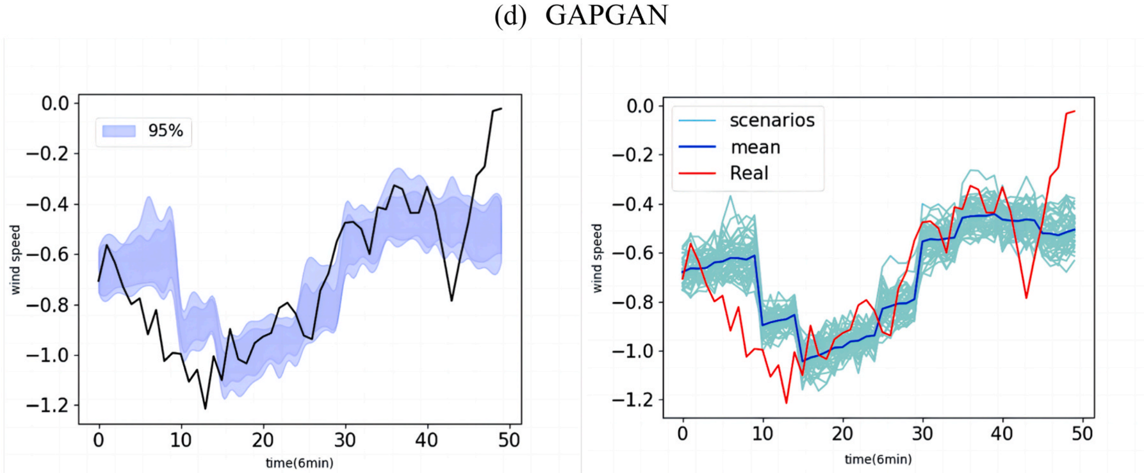
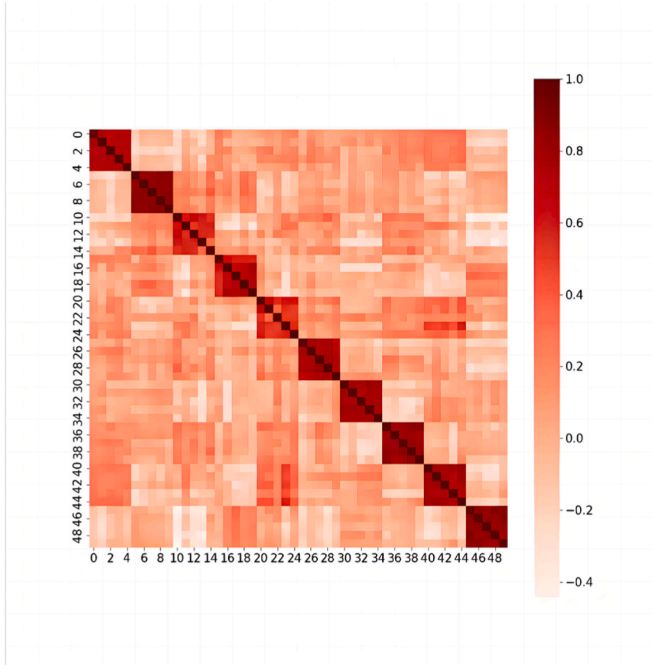


Fig. 15. (*continued*).

the reliability is decreased by 0.02–0.04 and the overall performance is undermined.

In Fig. 14, distributions of MSCP, PINAW and MSCWC for 11 wind turbines are visualized. Results in Fig. 14 (c) show that the CNN-GAN has an advantageous MSCWC overall, as evidenced by the lowest and tightest box. In contrast, Fig. 14 (a) and (b) strongly imply that CNN-GAN is in favor of reliability but neglects the sharpness of prediction intervals. Finally, distributions of MSCWC values for the most of wind turbines are similar, characterized as tight boxes, but with some special cases due to variations in MSCP and PINAW.

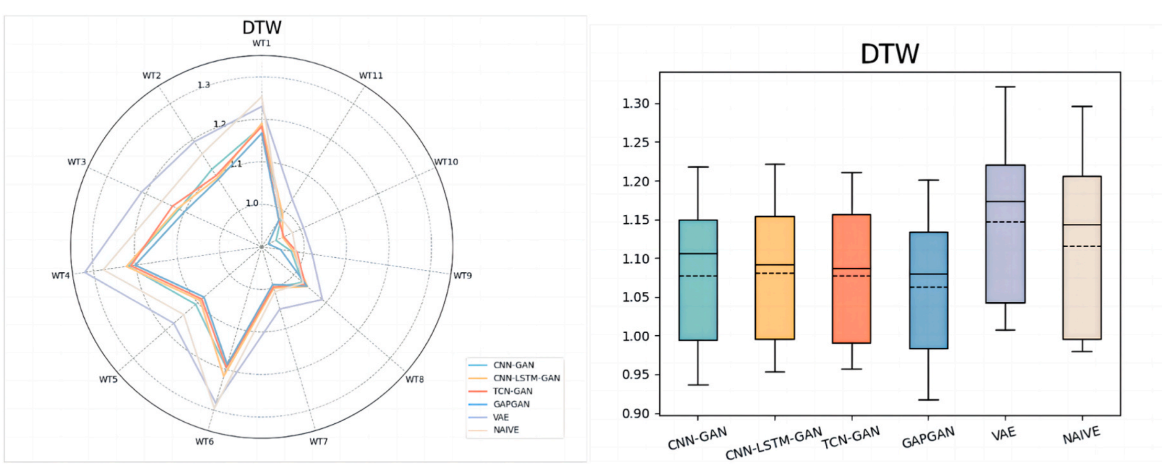


(e) VAE

Fig. 15. (*continued*).

Table 7
DTW for 30-min ahead wind speed predictions for all considered models.

	WT1	WT2	WT3	WT4	WT5	WT6	WT7	WT8	WT9	WT10	WT11	MEAN
CNN-GAN	1.183	1.117	1.114	1.218	1.106	1.196	1.000	1.027	0.970	0.937	0.988	1.078
CNN-LSTM-GAN	1.190	1.093	1.118	1.222	1.092	1.217	1.001	1.042	0.976	0.954	0.991	1.081
TCN-GAN	1.183	1.099	1.131	1.211	1.087	1.195	0.998	1.035	0.984	0.957	0.975	1.077
GAPGAN	**1.167**	**1.089**	**1.100**	**1.201**	**1.080**	**1.188**	**0.992**	1.041	**0.945**	**0.917**	**0.975**	**1.063**
VAE	1.230	1.194	1.211	1.321	1.174	1.283	1.052	1.089	1.020	1.008	1.033	1.147
NAIVE	1.253	1.160	1.159	1.277	1.143	1.296	1.010	**1.029**	0.981	0.980	0.982	1.115



(a) (b)

Fig. 16. (a) The radar map and (b) the box-whisker plot of DTW values for all models.

In addition to the uncertainty calibration of the generated scenarios, the short-term SG also needs to ensure the diversity of scenarios. In Fig. 15, the effect of different models on SG predictions for a certain period of time is shown in various forms, including the correlation heat maps among 50 scenarios generated by models. The heat maps in Fig. 15 imply that the scenarios generated by GAPGAN have less correlations

Table 8
The average training result comparison between C-Train, A-Train, and A-Train-O.

Training methods	MAE	RMSE	DTW	MSCWC	PINAW	MSCP
C-Train	0.225	0.259	1.079	0.255	0.062	0.458
A-Train	0.225	0.258	1.091	0.251	0.070	0.498
A-Train-O	0.220	0.253	0.975	0.236	0.071	0.520

and better diversity, proving that GAPGAN can effectively address the mode collapse problem.

5.3.3. *Shape similarity*

In this section, we compare and analyze the shape similarity between the main trend of the generated scenarios and real observations for different models using DTW, which is often used to measure the similarity between two time series of arbitrary lengths. The results of DTW for all models on the test set are listed in Table 7.

As shown in Table 7, the GAPGAN model proposed in this paper leads to the outstanding performance in generating scenarios of appropriate shapes. Except for the wind turbine 8, GAPGAN always achieves the lowest DTW values, indicating the outstanding shape similarity. To illustrate the robustness of the proposed models, distributions of DTW values for different models are also presented in Fig. 16 and the same conclusion can be obtained.

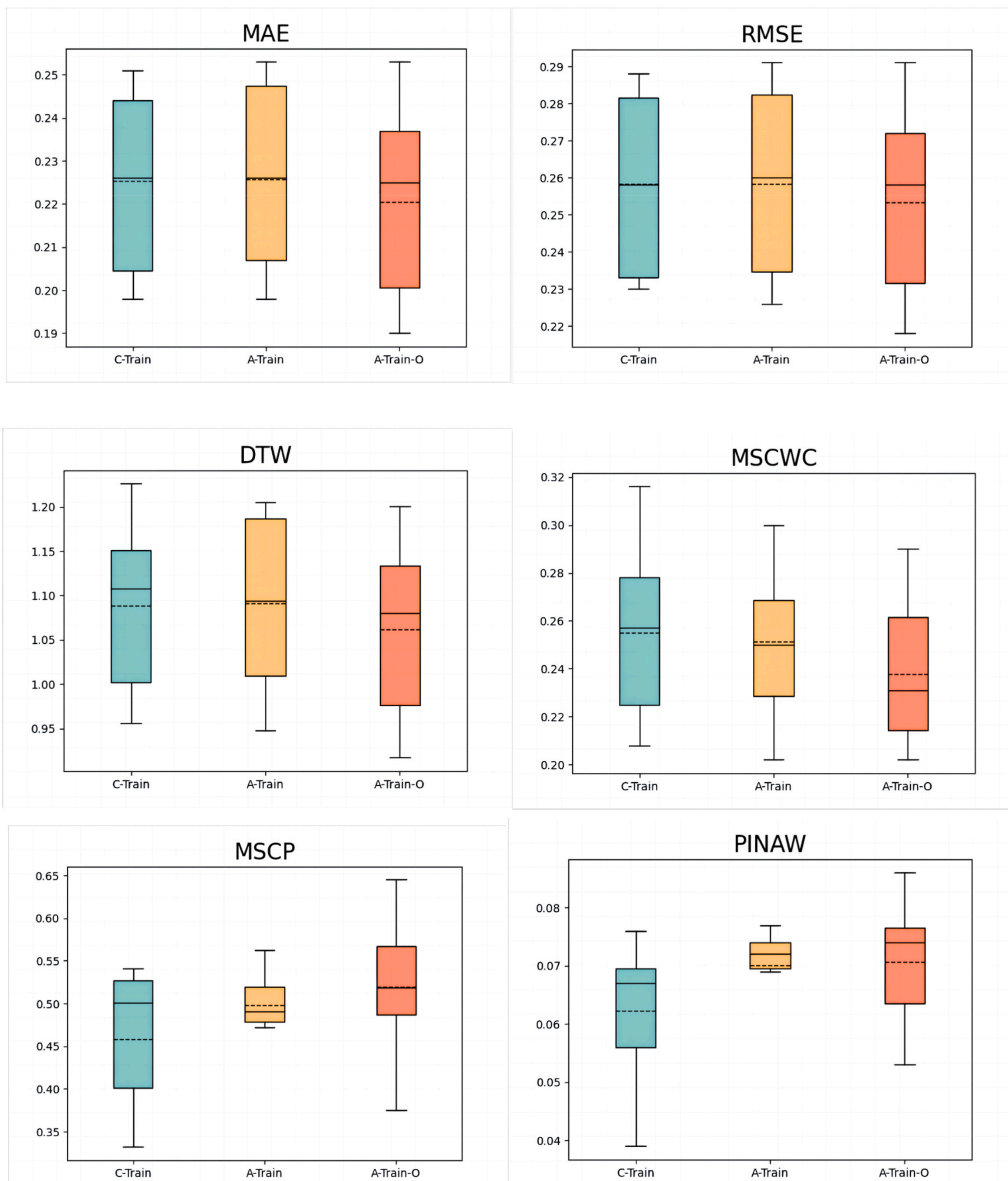


Fig. 17. Box-whisker plots of MAE, RMSE, DTW, MSCWC, MSCP and PINAW for three training methods.

5.3.4. Adaptive training effect

To evaluate the effectiveness of the proposed adaptive training and optimal model selection mechanisms, three GAN training methods are considered for an ablation study: Conventional Training (C-Train), Adaptive Training (A-Train), and Adaptive Training with Optimal model selection (A-Train-O). The C-Train refers to the framework in which both the generator and discriminator are updated in each training epoch. In the A-Train framework, the mechanism proposed in Section 4.3.2 for controlling the training epochs for the discriminator is added. In A-Train-O, the mechanisms proposed in Sections 4.3.2 and 4.3.3 are both implemented.

In Table 8, the average results of six evaluation metrics evaluated on the test set for 11 wind turbines are presented for three training methods. As shown in Table 8, three important conclusions can be drawn. First, compared to C-Train and A-Train, the A-Train-O method proposed in this paper demonstrates better performance in point prediction accuracy, shape similarity, and uncertainty evaluation accuracy. Second, there is little difference in point prediction accuracy between A-Train and C-Train, but C-Train outperforms A-Train in terms of better shape similarity. Third, C-Train results in narrower interval predictions but lower interval coverage rate, while A-Train provides more reasonable interval width and higher multi-point coverage rate, leading to improved quality of interval prediction as shown by its lower MSCWC value. This ablation study result suggests that using the adaptive training mechanism alone could improve the uncertainty calibration accuracy to some extent. To achieve a larger improvement, the optimal model selection mechanism should be incorporated.

Distributions of results for the six evaluation metrics for 11 wind turbines are presented in Fig. 17. It is found that the boxes of A-Train-O are likely to be more compact than others in MAE, RMSE and DTW; while regarding the MSCWC, MSCP and PINAW, although the boxes of A-Train could be narrower, A-Train-O is apparently overall better as indicated by its lower MSCWC values. Therefore, it can be concluded that the proposed A-Train-O has a better generalization ability for different wind turbines.

6. Conclusion

In this paper, a novel wind speed scenario prediction method, GAPGAN, was proposed for predicting the short-term future wind speed accurately and evaluating the uncertainty properly. In the proposed model, based on GNN and TCN models, data were properly structured and spatiotemporal features of the wind field were sufficiently extracted and exploited. Secondly, an adaptive predictive GAN framework was proposed to ensure the training balance between the generator and discriminator and improve the model training performance. Thirdly, to accommodate multiple functions of scenario generations, a novel evaluation system for scenario generation was proposed, which could effectively facilitate the selection of models with the best comprehensive performance from the training process.

Computational studies were conducted based on SCADA data from 11 wind turbines in a wind farm. The effectiveness of the proposed GAPGAN method was verified by comparing it with five benchmarking models. The results of MAE, RMSE and DTW suggested that the proposed GAPGAN model could produce wind speed scenarios with better diversity, higher accuracy and more authentic shapes than the other five models in general. Besides, it was also found that generated scenarios can be converted to both reliable and moderately sharp wind speed interval prediction results. Finally, the proposed adaptive training mechanism was proved to be more effective than the conventional adversarial training scheme.

In future studies, we plan to investigate the wind speed scenario generation method for a longer prediction horizon, such as up to 24 h, and explore more effective feature extraction mechanism accordingly. Also, the value of the proposed scenario generation method for downstream wind farm management applications, such as a stochastic control model of wind turbines, should be examined.

CRediT authorship contribution statement

Xin Liu: Writing – original draft, Supervision, Project administration, Methodology, Conceptualization. **Jingjia Yu:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Lin Gong:** Supervision, Resources, Funding acquisition. **Minxia Liu:** Supervision, Project administration, Formal analysis. **Xi Xiang:** Supervision, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Hyperparameters of the GAPGAN model

Hyperparameters/layers	Values
Training params	Learn_rate = 1e-4, dropout_ratio = 0.1, optimizer = RMSprop
Graph learning layer	Embedding layer(num_node = 27, node_dim = 40) Linear(num_dim = 40, node_dim = 40)
Graph convolutional module	Number_of_modules = 3, gcn_depth = 3
Temporal convolutional module	Number_of_modules = 3, dilation=(1, 2), number_of_kernels = 32
Residual Connection	Number_of_layers = 3, number_of_kernels = 32
Skip Connection	Number_of_layers = 3, number_of_kernels = 64
Output Module	output_dim = 40

Appendix B. Algorithm 1. APGAN training algorithm

Algorithm 1. APGAN training algorithm

1: Input: Training set $\{D_{\text{train}}, P_{\text{train}}\}$, generator $g(\bullet)$ with initialized parameter θ, discriminator $d(\bullet)$ with initialized parameter φ;
2: Define: $G_{\text{tmp}} = 0$, $G_{\text{loss\_post}} = 0$, count $= 0$, learning rate: α, the number of training epochs: T, the number of iterations in each training epoch: N, batch size: B, hyperparametes n and m.
3: for $iter = 1$ to T do
4: if $G_{tmp} > 0$ then
5: for $k = 1$ to N do
6: draw B samples from $\{D_{\text{train}}, P_{\text{train}}\}$ and draw B random noise data $\{Z\}$ from $N(0, I)$;
7: compute and record the loss value of the generator $\mathscr{L}_G(\theta)$, update parameters: $\theta \leftarrow \theta - \alpha \cdot \nabla_\theta \mathscr{L}_G(\theta)$;
8: end for
9: $G_{tmp} \leftarrow G_{tmp} - 1$;
10: else
11: for $k = 1$ to N do
12: draw B samples from $\{D_{\text{train}}, P_{\text{train}}\}$ and draw B random noise data $\{Z\}$ from $N(0, I)$;
13: compute the loss of the generator $\mathscr{L}_G(\theta)$, update parameters: $\theta \leftarrow \theta - \alpha \cdot \nabla_\theta \mathscr{L}_G(\theta)$;
14: compute the loss of the discriminator $\mathscr{L}_D(\varphi)$, update parameters: $\varphi \leftarrow \varphi - \alpha \cdot \nabla_\varphi \mathscr{L}_D(\varphi)$;
15: end for
16: compute the mean value of $\mathscr{L}_G(\theta)$ in the last epoch as G_{loss};
17: if $G_{\text{loss}} > G_{\text{loss\_post}}$
18: count$\leftarrow$count $+ 1$;
19: else
20: count $\leftarrow 0$;
21: end if
22: $G_{\text{loss\_post}} \leftarrow G_{\text{loss}}$;
23: if count $\geq n$ then
24: $G_{\text{tmp}} \leftarrow m$;
25: end if
26: end if
27: end for

Appendix C. Algorithm 2. Optimal model selection

Algorithm 2 Optimal model selection

1: Input: Training set $\{D_{\text{train}}, P_{\text{train}}\}$, validation set $\{D_{\text{val}}, P_{\text{val}}\}$
2: Set: $E = 0$, the number of iterations to start saving the model: s, the length of epoch intervals to evaluate and save the generator L; the number of training epochs: T; the size of validation set N_{val}; hyperparameters $\alpha_1, \alpha_2, \alpha_3$.
3: for $iter = 1$ to T do
4: train the generator and discriminator according to lines 4–25 in Algorithm 1;
5: if $iter \geq s$ and $(iter - s)$ % $L = 0$
6: generate a set of predicted scenarios denoted by $\{\widehat{P}_{val}{}^{(i)}\}, i = 1, 2, \ldots, N_{val}$;
7: compute the mean value of scenarios denoted by $\{\widehat{P}_{mean}{}^{(i)}\}, i = 1, 2, \ldots, N_{val}$;
8 $eval^{(i)} = \alpha_1 \bullet RMSE(P_{val}{}^{(i)}, \widehat{P}_{mean}{}^{(i)}) + \alpha_2 \bullet DTW(P_{val}{}^{(i)}, \widehat{P}_{mean}{}^{(i)}) + \alpha_3 \bullet MSCWC(P_{val}{}^{(i)}, \widehat{P}_{val}{}^{(i)}), i = 1, 2, \ldots, N_{val}$;
9: $eval = \dfrac{1}{N_{val}} \sum_{i=1}^{N_{val}} eval^{(i)}$;
10: if $iter = s$
11: save the generator model G;
12: $E \leftarrow eval$;
13: else:
14: if $eval < E$ then
15: $E \leftarrow eval$;
16: update the generator model G;
17: end if
18: end if
19: end for

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